

Ray Pomponio, MS

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Professional Summary

Biomedical data scientist with 7+ years' experience modeling complex data and longitudinal outcomes. Collaborates with senior investigators to assess feasibility and deliver timely, impactful analyses for publication. Currently leads analytics for 3 studies using pediatric public health surveillance data, located in Pittsburgh, PA.

Highlighted Project Experience

Lead Biostatistician for Study of Viral Circulation in New Vaccine Surveillance Network (NVSN)

September 2024 – Present

- Performed multivariate statistical analysis to assess illness severity, adjusted for suspected correlates, risk factors, and viral-genetic subtypes in children with human Metapneumovirus (MPV).
- Wrote statistical sections of manuscript submitted to scientific journal, received peer review (under revision).

Design of FDA Protocol for Medical Device Trial

March 2024 – June 2024

- Conducted sample size and power calculations in R for medical device study based on non-inferiority targets.
- Developed statistical analysis plan for FDA Q-submission.

Randomization Scheme for R01 Study in Pulmonology

August 2021 – September 2022

- Designed randomization protocol for study of aspiration in intubated patients, utilizing permuted block schema.

Work Experience

Staff Data Scientist

Children's Hospital of Pittsburgh, Division of Infectious Diseases – Pittsburgh, PA

May 2024 – Present

- Conduct statistical analyses with biomedical datasets to support pediatric public health initiatives.
- Collaborate with cross-functional teams to design clinical studies and interpret data.
- Develop statistical protocols for NIH grant submissions, performing power calculations and sample justifications.
- Present findings to staff and faculty, contributing to publications and medical conferences on infectious disease.

Graduate Research Assistant

Colorado School of Public Health, Department of Biostatistics – Denver, CO

August 2021 – December 2023

- Managed multiple ongoing data analytics projects focused on pulmonology and critical care health outcomes.
- Applied statistical methods and R/Python programming to clean, analyze, and visualize data.
- Co-authored 7+ peer-reviewed publications and presented at industry conferences.

Neuroimaging Data Analyst

University of Pennsylvania, Department of Radiology – Philadelphia, PA

March 2018 – August 2020

- Analyzed high-dimensional imaging data to support research in radiology and neuroscience.
- Built Python package to streamline combination of diverse neuroimaging datasets with clinical information.
- Collaborated with radiologists and neurologists to identify lifespan trends and interpret results.
- Contributed to the development of predictive models of age and dementia for clinical diagnostics.

Education

Doctoral-level coursework: Biomedical Machine Learning, Model Selection, Causal Inference (current).

Master of Science in Biostatistics

University of Colorado Anschutz Medical Campus – Aurora, CO

Graduated: May 2023 (4.0 GPA)

Marvin Porter Award for outstanding academic achievement

Bachelor of Science in Economics

University of Pennsylvania, Wharton School – Philadelphia, PA

Graduated: May 2018 (Magna cum laude)

Skills

- Statistical Programming: R, SAS, Python
- Data Manipulation & Analysis: Regression, longitudinal modeling, survival analysis, causal inference
- Scripting & Automation: Unix Bash, Microsoft Power Automate
- Data Visualization: ggplot2, shiny, matplotlib, Power BI
- Soft Skills: Project management, Request triage, Statistical consulting, Scientific communication

Selected Publications

Pomponio R, Peterson RA, Owusu M, Slaughter S, Melgar S, Jolley SE, Burnham EL. Phosphatidylethanol measures in patients with severe COVID-19-associated respiratory failure identify a subset with alcohol misuse. *Alcohol Clin Exp Res*. 2025 Jan;49(1):165-174.

Held N, Neumeier A, Amass T, Harry E, **Pomponio R**, Peterson RA, Huie TJ, Moss M. Extraneous Load, Patient Census, and Patient Acuity Correlate with Cognitive Load During ICU Rounds. *Chest*. 2024 Jun;165(6):1448-1457.

Burnham EL, **Pomponio R**, Perry G, Offner PJ, Ormesher R, Peterson RA, Jolley SE. Prevalence of Alcohol Use Characterized by Phosphatidylethanol in Patients with Respiratory Failure Before and During the COVID-19 Pandemic. *CHEST Crit Care*. 2024 Mar;2(1).

Nasrallah IM, Gaussoin SA, **Pomponio R**, Dolui S, Erus G, Wright CB, Launer LJ, Detre JA, Wolk DA, Davatzikos C, Williamson JD, Pajewski NM, Bryan RN. Association of Intensive vs Standard Blood Pressure Control with Magnetic Resonance Imaging Biomarkers of Alzheimer Disease: Secondary Analysis of the SPRINT MIND Randomized Trial. *JAMA Neurol*. 2021 May 1;78(5):568-577.

Pomponio R, Erus G, Habes M, Doshi J, Srinivasan D, Mamourian E, Bashyam V, Nasrallah IM, Satterthwaite TD, Fan Y, Launer LJ, Masters CL, Maruff P, Zhuo C, Völzke H, Johnson SC, Fripp J, Koutsouleris N, Wolf DH, Gur R, Gur R, Morris J, Albert MS, Grabe HJ, Resnick SM, Bryan RN, Wolk DA, Shinohara RT, Shou H, Davatzikos C. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *Neuroimage*. 2020 Mar; 208:116450.

Complete List of Published Work in MyBibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/raymond.pomponio.1/bibliography/public/>

Other Relevant Experience

- Maintainer of 'neuroHarmonize' python package (2021-present): pypi.org/project/neuroHarmonize/
- Statistical reviewer for five journal articles (2020-present): *Imaging Neuroscience*, *NeuroImage*
- Graduate teaching assistant (Fall 2023): *Analysis of Longitudinal and Survival Data*
- Volunteer: *Humane Animal Rescue of Pittsburgh* (40+ hours), *First Tee of Pittsburgh* (20+ hours)